28/07/2025

22AI501 - AI

Discourse - PT - I

1. 4 Invalid statements → (2), (3), (5), (7).

2. a) Offline Search.
   b) Local Search.
   c) Exploration Strategy
   d) Online Search.

3. A1 - B3 - C2
   A2 - B4 - C3
   A3 - B2 - C4
   A4 - B1 - C1

4. Four Invalid Statements → (b), (d), (g), (h).

5. a) Both A and R are true and R is correct exp. of A.

6. a) $O(b^d)$
   b) $O(b.m)$
   c) $O(b^l)$
   d) $O(b^{(1+c/\epsilon)})$*
   e) $O(l)$ (or) $O(b.l)$

7. Step 4 → Constraint checking.
   Step 5 → Backtrack/Undo Assignment

8. Local Search
   A → global maximum.
   B → "flat" Local Maximum.
   C → current state.
   D → shoulder

9) a) (Hint : 3)
b) (Hint : 1)
c) (Hint : 2)
d) (Hint : 5)
e) (Hint : 4)
Step 4 : (Hint : 6)

10) 4 Invalid statements → (b), (d), (e), (f).

11) Given statement is True,
Justification: This process gradually improves the solution exactly as described.

12) Four Invalid → (b), (e), (f), (g) (or) (a), (b), (e), (f).

13) Step 3 : Ontology analysis
5 : Query and Reasoning mechanisms
6 : Integeration with Systems

14) Unification rule fails → Object cannot be identified as Mango due to color mismatch.

15) a) Rearrange steps → (5), (1), (3), (6), (2) (4).
b) False. (Analog circuit is valid & appropriate application)

16) First Order Logic Formula:

$$\forall x \, \exists y \, (\text{Parent}(y, x) \rightarrow \text{Parent}(x, y))$$

Expansion: For every 'x', there exists a 'y' such that if 'x' is a Parent of 'y', then 'y' is a Parent of 'x'.

17) In short-hand notation,

$$\exists x : boys(x) \wedge intelligent(x)$$

a) Some boys play cricket:

$$\exists x : Boys(x) \wedge Plays\ cricket(x)$$

b) Only one student failed in Maths.

$$\exists x\ (Student(x) \wedge failed\ in\ Math(x) \wedge \forall y\ ((Student(y) \wedge failed\ in\ Math(y)) \rightarrow (x = y)))$$

18) a) UnStack (A, B)
c) PickUP (C)
d) Stack (C, A)
f) Stack (A, B)
g) Done (all blocks are in desired positions)
h) Hand Empty.